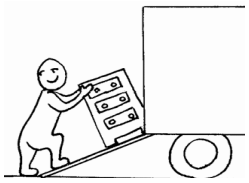


Inclined Planes Mini-Assessment



Use Pearson Science p.301 - 305 TOTAL: _____ 39

1. **List** three places you have used a ramp.

   a. ____ **trailer, road, shopping centre, any ramp!**

   b. _____

   c. _____

   (3 marks)

2. **List** three common examples of a wedge.

   a. _____ **axe, incisor teeth, pins, needles, doorstops, nails**

   b. _____

   c. _____

   (3 marks)

3. **State** whether the following are true or false.

   a. The longer the slope of a ramp for a certain height, the greater its mechanical advantage.

      _____ **T** _____

   b. A screw with a small pitch corresponds to a steep inclined plane.

_____ **F** _____

c. A screw with a small pitch requires fewer turns to be inserted into a piece of wood than a screw with a large pitch.

_____ **F** _____

d. A wedge changes the direction of a force by 90° and increases its size.

_____ **T** _____ (4 marks)

4. **State** an example of a type of screw that cuts through a:

a) solid _____ **wood screw**

b) liquid _____ **boat propeller**

c) gas _____ **aircraft propeller**

(3 marks)

5. **Recall** five devices that can be produced by flaking.

_____ **spearhead, chisels, axes, saws, knives**

_____
_____

_____
_____

(5 marks)

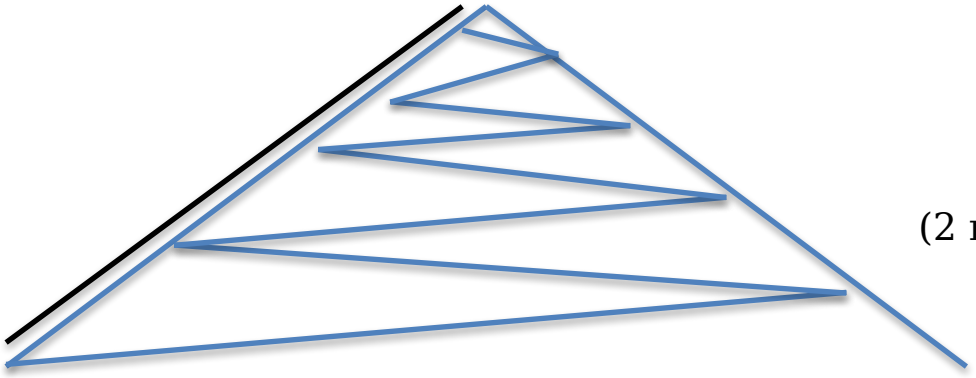
6. **Recall** one name for a spear thrower _____ **woomera**

(1 mark)

7. You have been given the job of building a road up the following hill.

a) **Draw** in the road that would be shortest but hardest to walk up.

b) **Draw** in a road that would be long but easy to walk up.



(2 marks)

8. **Describe** how a stone is flaked

_____

A block of stone (called a core), is struck with a hammer stone

(usually a pebble), to chip off a sharp piece of stone, called a flake.

(2 marks)

9. **Explain** why stone rich in silica is suitable to use in creating stone tools

They are hard but brittle.

_____
_____

_____
_____

(1 mark)

10. **Identify** which type of simple machine (inclined plane, wedge or screw) is described by these devices:

a) corkscrew _____ **screw**

b) axe _____ **wedge**

c) electric fan _____ **screw**

d) car park ramp _____ **inclined plane**

e) chisel _____ **wedge**

f) escalator _____ **inclined plane**

(6 marks)

11. Mechanical advantage is equal to $\frac{\text{length of slope}}{\text{height of ramp}}$

Calculate the mechanical advantage of the following ramps:

Length of Slope (m)	**Height of Ramp (m)**	**Mechanical Advantage**
15	3	**5**
50	25	**2**
12	2	**6**
20	5	**4**
240	30	**8**

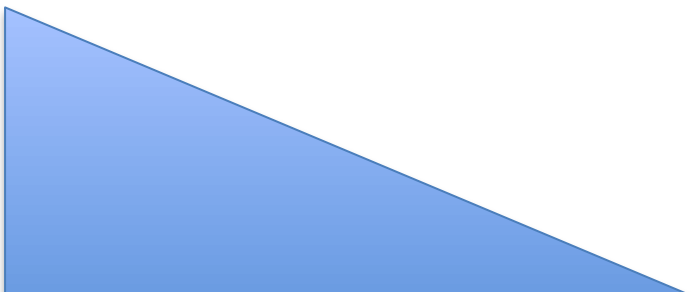
(5 marks)

12. Use a piece of scrap paper to cut out a gentle slope and a steep slope (see p.304 SB) and wrap them around a pencil.

Gentle Slope



Steep Slope



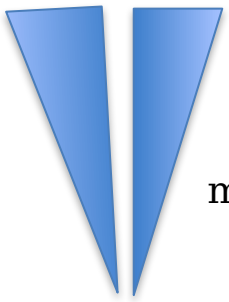
When you wrapped them around a pencil and produced a "screw", was the thread spread out or tight together?

a) Gentle slope _____ **tight**

b) Steep slope _____ **spread**

(2 marks)

13. **Draw** an axe head and show how it is two inclined planes.



(2 marks)